

Phase 8:Project Demonstration

SmartLenderAI-Powered Credit Card Approval System

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Demo Scenario 1 – Automated Screening

A credit analyst enters an applicant's financial profile into the web application. The model returns an instant approve/reject prediction, letting the analyst prioritize which applications need closer review.

Demo Scenario 2 – High-Risk Identification

A compliance officer screens applicants with past-due loan records. The feature engineering pipeline converts multi-class payment status into a binary risk label, so high-risk applicants are clearly flagged.

Demo Scenario 3 – Self-Service Eligibility Check

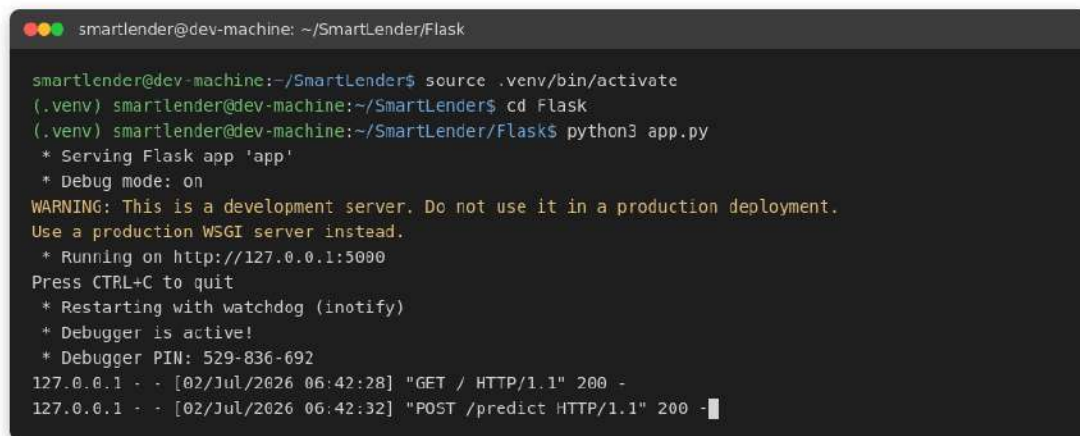
A prospective customer enters their own details before formally applying and receives an instant likelihood of approval, reducing the number of applications that would otherwise be rejected outright.

Conclusion & Future Scope

SmartLender shows that a tuned XGBoost model can automate a meaningful share of the credit approval workload. Future work includes adding explainability for each decision and monitoring the model for drift as new applicant data comes in.

Terminal Execution

The screenshot below shows the Flask development server starting successfully and serving requests on local port 5000.

A terminal window with a dark background and light green text. The prompt is 'smartlender@dev-machine: ~/SmartLender/Flask'. The user enters 'source .venv/bin/activate', then 'cd Flask', and finally 'python3 app.py'. The output shows the server starting, with a warning about production use, the address 'http://127.0.0.1:5000', and a debugger PIN. Two HTTP requests are shown: a GET request to '/' and a POST request to '/predict', both returning a 200 status.

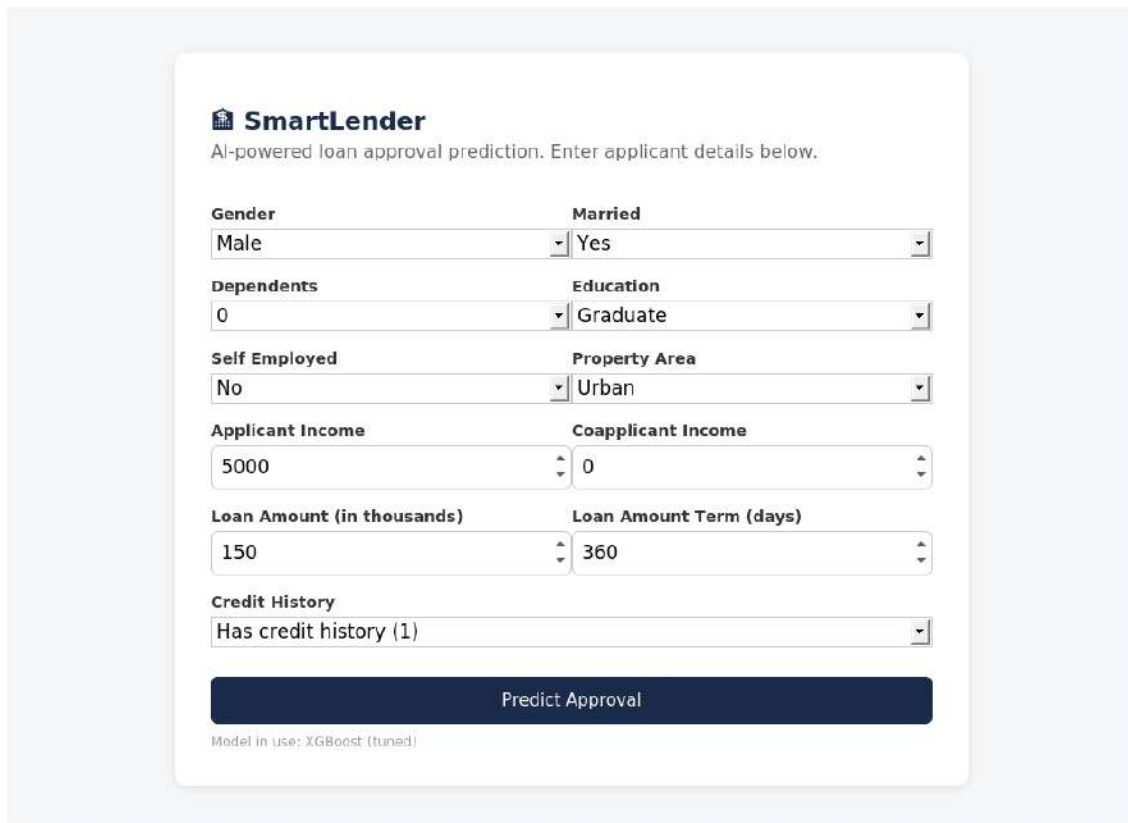
```
smartlender@dev-machine: ~/SmartLender/Flask

smartlender@dev-machine:~/SmartLender$ source .venv/bin/activate
(.venv) smartlender@dev-machine:~/SmartLender$ cd Flask
(.venv) smartlender@dev-machine:~/SmartLender/Flask$ python3 app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (inotify)
* Debugger is active!
* Debugger PIN: 529-836-692
127.0.0.1 - - [02/Jul/2026 06:42:28] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Jul/2026 06:42:32] "POST /predict HTTP/1.1" 200 -
```

Figure 5: Flask server running in terminal

Web Interface – Input Form

The applicant enters demographic and financial details through the web form shown below.



The image shows a web interface for 'SmartLender', an AI-powered loan approval prediction tool. The form is titled 'SmartLender' with a house icon and the subtitle 'AI-powered loan approval prediction. Enter applicant details below.' The form contains several input fields arranged in a grid-like fashion. The fields are: 'Gender' (dropdown with 'Male' selected), 'Married' (dropdown with 'Yes' selected), 'Dependents' (dropdown with '0' selected), 'Education' (dropdown with 'Graduate' selected), 'Self Employed' (dropdown with 'No' selected), 'Property Area' (dropdown with 'Urban' selected), 'Applicant Income' (spinbox with '5000' entered), 'Coapplicant Income' (spinbox with '0' entered), 'Loan Amount (in thousands)' (spinbox with '150' entered), 'Loan Amount Term (days)' (spinbox with '360' entered), and 'Credit History' (dropdown with 'Has credit history (1)' selected). At the bottom of the form is a large dark blue button labeled 'Predict Approval'. Below the button, in small text, it says 'Model in use: XGBoost (tuned)'.

Field	Value
Gender	Male
Married	Yes
Dependents	0
Education	Graduate
Self Employed	No
Property Area	Urban
Applicant Income	5000
Coapplicant Income	0
Loan Amount (in thousands)	150
Loan Amount Term (days)	360
Credit History	Has credit history (1)

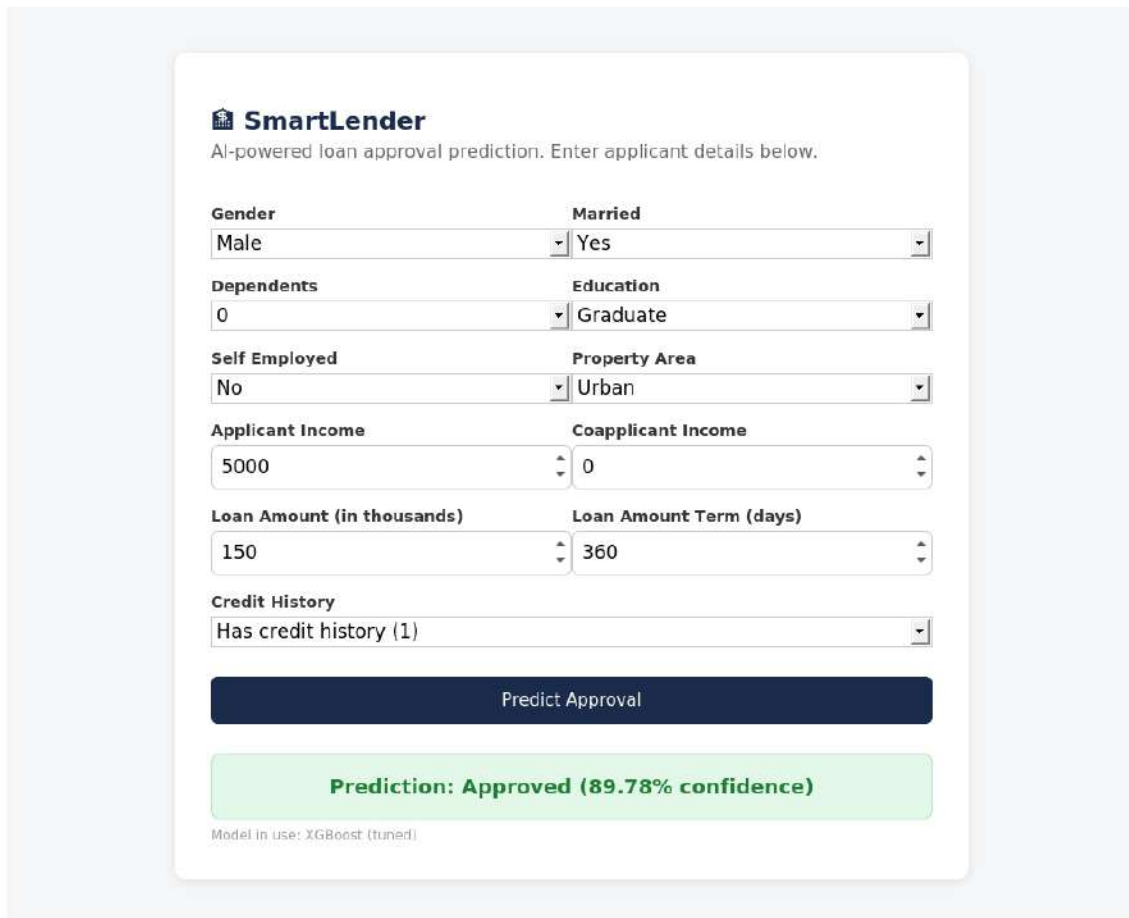
Predict Approval

Model in use: XGBoost (tuned)

Figure 6: Input form interface

Web Interface – Prediction Output

After submission, the system returns the approve/reject decision along with the tuned XGBoost model's confidence percentage.



The image shows a web interface for 'SmartLender', an AI-powered loan approval prediction system. The interface is titled 'SmartLender' with a house icon and the subtitle 'AI-powered loan approval prediction. Enter applicant details below.' It features a form with various input fields for applicant details, a 'Predict Approval' button, and a green box displaying the prediction result: 'Prediction: Approved (89.78% confidence)'. Below the prediction, it states 'Model in use: XGBoost (tuned)'.

SmartLender
AI-powered loan approval prediction. Enter applicant details below.

Gender	Married
Male	Yes
Dependents	Education
0	Graduate
Self Employed	Property Area
No	Urban
Applicant Income	Coapplicant Income
5000	0
Loan Amount (in thousands)	Loan Amount Term (days)
150	360
Credit History	
Has credit history (1)	

Predict Approval

Prediction: Approved (89.78% confidence)

Model in use: XGBoost (tuned)

Figure 7: Prediction output display